

BLACK SMOKERS

Ocean water seeping into the rift zone is heated by contact with the hot rock, but high pressure stops it from boiling. It gets hotter and hotter, reaching 750°F (400°C)—four times higher than its normal boiling point. The very hot water dissolves chemicals in the rock, and eventually this chemical-rich water is forced back up into the ocean. When it hits the cold ocean water, the chemicals form dark particles that look like smoke billowing from the rift, so these plumes are known as black smokers.

WINDING RIDGES

The mid-ocean ridges, along with their black smokers and underwater volcanoes, form a network that extends through all the oceans of the world. They form the divergent boundaries where the giant plates of Earth's crust are moving apart. Convergent boundaries mark where the plates are pushing together.



- KEY**
- Divergent boundary
 - Convergent boundary
 - Sliding boundary



INSIDE A BLACK SMOKER

As the chemicals in the water erupting from black smokers turn solid, they build up around the plumes of hot water to form rocky, chimneylike structures. These can be up to 98 ft (30 m) high.

A mineral chimney can grow 12 in (30 cm) a day

Plume is often black but can be white

